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ECE 414 Senior Design Project Proposal

Team Number: 9

Project Title: Multilingual Sign-to-Speech Communication Device: Edge-Deployed ASL Recognition with Confidence-Gated Speech Output

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8. Relevant Technical Background

This file is Section 8 of the ECE 414 proposal draft, submitted on its own. Section, figure, table and reference numbers are those of the full proposal (docs/ECE_414_Proposal_Draft_with_Section8.docx); the reference list below keeps only the entries cited here.

This section reviews in detail the two research articles and the four patents that are closest to the proposed device, and explains for each how our proposal differs from it. Articles were found with Google Scholar (https://scholar.google.com) and patents with Google Patents (https://patents.google.com), using combinations of the terms sign language recognition, sign language translation, isolated sign, pose, camera, speech and confidence. Patents were selected for being close to our design and for representing the main approaches: a single camera with cloud processing (Sign-Speak), multiple cameras with a depth sensor (Avodah), translation inside a video relay service (Sorenson) and sensor gloves (Bosch). For every patent the abstract is quoted from Google Patents, and the legal status was checked there on 26 October 2026. For the articles, the identifiers (DOI, ISBN and pages) were confirmed with the publishers' records. Section 8.4 compares all of them with the proposed device.

8.1 Article 1: ASL Citizen Dataset

Table 17. Bibliographic record of Article 1.

Item	Details
Title	ASL Citizen: A Community-Sourced Dataset for Advancing Isolated Sign Language Recognition
Authors	Aashaka Desai, Lauren Berger, Fyodor O. Minakov, Vanessa Milan, Chinmay Singh, Kriston Pumphrey, Richard E. Ladner, Hal Daumé III, Alex X. Lu, Naomi Caselli, Danielle Bragg
Published in	Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track, pp. 76893–76907, 2023 [2]
ISBN / ISSN	ISBN 978-1-7138-9911-2 (proceedings volume 36); ISSN 1049-5258
DOI	10.52202/075280-3360 (proceedings); 10.48550/arXiv.2304.05934 (preprint)
Links	https://doi.org/10.52202/075280-3360; https://arxiv.org/abs/2304.05934; dataset: https://www.microsoft.com/en-us/research/project/asl-citizen/

Description. The article introduces ASL Citizen, the first crowdsourced isolated sign language recognition dataset collected with the consent of its participants. It contains 83,399 videos of 2,731 distinct ASL signs, recorded by 52 Deaf and hard-of-hearing signers with their own

webcams in everyday environments, about 30 videos per sign. Recruitment and consent materials were presented in ASL first, and participants were compensated. The authors frame isolated sign recognition as dictionary retrieval: a user demonstrates a sign to a webcam and the system returns a ranked list of matching signs. The 52 signers are divided into fixed training, validation and test sets so that test signers are never seen in training, and results are reported as recall at 1, 5 and 10 (recall at 1 is ordinary accuracy), discounted cumulative gain and mean reciprocal rank.

Two baselines are trained. An appearance model, I3D, a three-dimensional convolutional network applied to the video frames, reaches 63.10% top-1, 86.09% top-5 and 90.86% top-10 accuracy on the unseen test signers. A pose model, ST-GCN, which works on MediaPipe Holistic keypoints, reaches 59.52% top-1. The authors also show that the source of the training data matters more than the model: on the signs shared with the older WLASL dataset, a model trained on ASL Citizen reaches 74.16% top-1 against 8.49% for the same model trained on WLASL. The data may be used for non-commercial research only.

Difference from our proposal and how we use it. ASL Citizen is a dataset and benchmark, not a device, so it is a resource for our project rather than a competing design. It is the main training and test data of our recognizer (Section 7.5): our v5 model, fine-tuned on a 134-word service-counter subset of it, reaches 86.6% top-1 and 98.2% top-5 on the same unseen test signers. The two results are not directly comparable, because the article ranks all 2,731 signs while our device chooses among 134 words. Beyond the dataset, our proposal adds what a deployed device needs and the article does not address: a small pose-based model that runs offline on embedded hardware (1.78 million parameters, about 4 ms per sign on one processor core), calibrated confidence with abstention and risk–coverage evaluation, accuracy reported for each signer, and the language and speech path from recognized signs to a spoken sentence. Its use case, dictionary look-up, is also different from ours, a conversation at a service counter.

8.2 Article 2: U.S. Deaf Community Perspectives on Automatic Sign Language Translation

Table 18. Bibliographic record of Article 2.

Item	Details
Title	U.S. Deaf Community Perspectives on Automatic Sign Language Translation
Authors	Nina Tran, Richard E. Ladner, Danielle Bragg
Published in	Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '23), New York, NY, USA, October 22–25, 2023, pp. 1–7 [1]
ISBN	979-8-4007-0220-4 (ASSETS '23 proceedings, ACM)
DOI	10.1145/3597638.3614507
Links	https://dl.acm.org/doi/10.1145/3597638.3614507; https://www.microsoft.com/en-us/research/publication/u-s-deaf-community-perspectives-on-automatic-sign-language-translation/

Description. The article reports a survey of 32 Deaf and hard-of-hearing ASL users in the United States about automatic sign language translation. Participants completed an ASL check, and ASL was the primary language of 81% of them. For 14 everyday settings they rated how often they had wanted an interpreter but could not get one, and how willing they would be to use automatic translation. The unmet need was largest in medical, professional and educational settings, while willingness was highest for going into a business, followed by self-service kiosks and informal education; attitudes toward medical use were sharply divided. Participants expected the highest accuracy in medical settings and the lowest for kiosk-type exchanges. On form factors, 91% were comfortable with a phone app, 69% with a stand-alone system such as a computer on a counter, 38% with a wearable and 6% with an implant.

The survey also measured concerns. More than 80% of participants were concerned about hearing people profiting from ASL, systems missing ASL grammatical features, automatic translation weakening the legal right to other accommodations, poorer performance for some users than for others, and limited Deaf involvement in leadership. The most valued benefit was access in more places and at the last minute. The authors conclude that developers should target specific settings chosen with the community and should involve Deaf people throughout.

Difference from our proposal and how we use it. This article is a user study, not a technical system, and it is the main evidence behind several of our design decisions (Sections 3.3 and 5). Following its results, our device targets the business and service-counter setting, where

willingness is highest and accuracy expectations are lower; it is a stand-alone counter device, the form factor 69% of participants accepted, rather than a wearable; it reports accuracy for each signer; and it is presented as an aid when no interpreter is available rather than as a replacement. The article goes further than our device in one respect: 91% of participants were concerned about systems that miss ASL grammar, and our device recognizes individual signs without facial grammar. We state this limitation openly instead of claiming full translation.

8.3 Patents

Four patents are reviewed in detail below, from the closest to our design to the most distant. Fig. 11 reproduces one drawing from each of the three closest. An expired Hitachi patent [21] is summarized at the end for completeness.

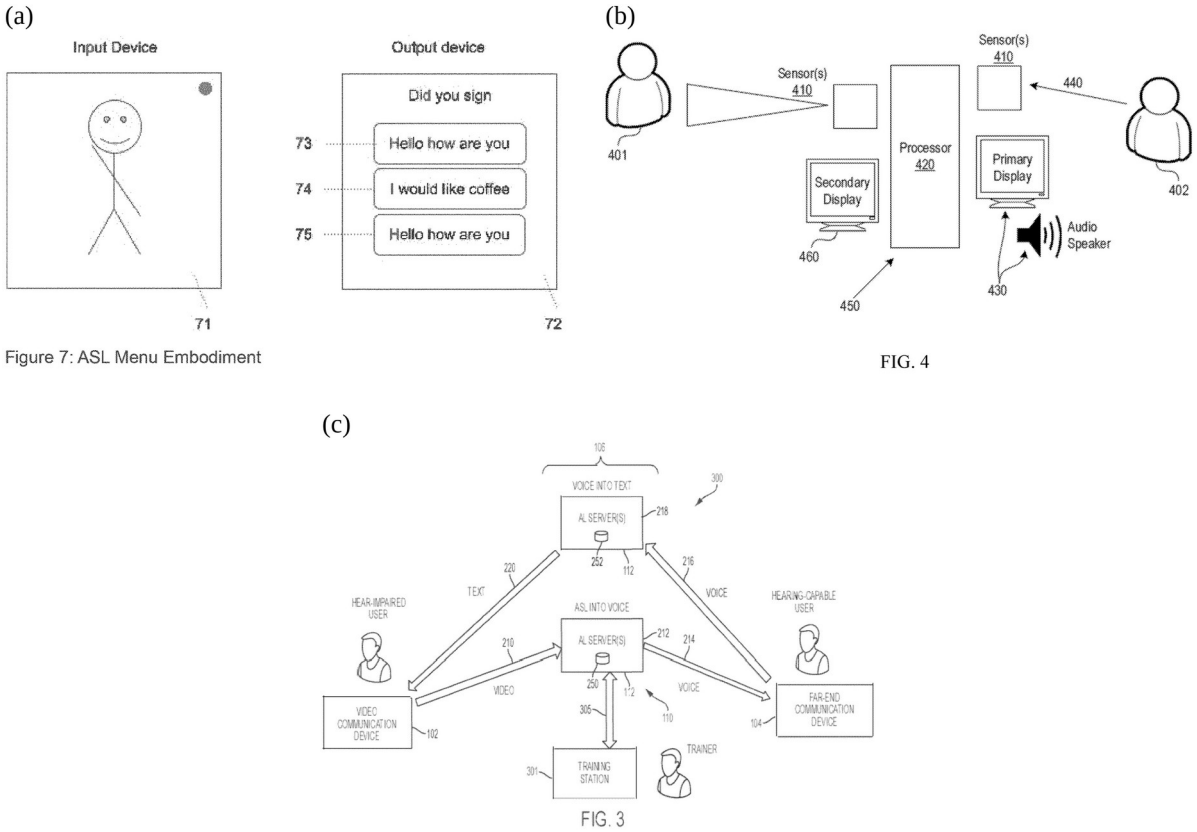


Fig. 11. Drawings reproduced from the reviewed patents. (a) Sign-Speak, FIG. 7: after the user signs, the three most likely sentences are shown for confirmation [19]. (b) Avodah, FIG. 4: one device between two people, with sensors, two displays and a speaker [18]. (c) Sorenson, FIG. 3: a video relay service in which artificial intelligence (AI) servers turn ASL video into voice and voice into text, with a training station [39].

Patent 1: Realtime AI Sign Language Recognition (Sign-Speak)

Table 19. Record of Patent 1.

Item	Details
Title	Realtime AI sign language recognition
Number	U.S. Patent No. 12,518,653 B2, granted January 6, 2026 [19]; published as application US 2022/0327961 A1 on October 13, 2022
Inventor / assignee	Nikolas Anthony Kelly / Sign-Speak Inc.
Dates	Priority May 11, 2020; filed May 11, 2021
Legal status	Active; Google Patents gives January 23, 2043 as the expected expiration
Link	https://patents.google.com/patent/US12518653B2/en

Abstract. "A real time sign language recognition method that allows Deaf and Hard of Hearing individuals to sign into any apparatus with a camera to extract target information (such as a translation in a target language) is proposed."

Description. The patent addresses conversations between Deaf signers and non-signers in impromptu settings where an interpreter cannot be provided immediately. It criticizes earlier systems that need gloves, 3D cameras or camera arrays, per-user calibration, or heavy computation on the device, and instead requires "only a device with internet connection and a single lens camera". Its general architecture has four stages, a capture device, a feature extractor, a processing module and an output device, and at least the feature extractor or the processing module must run, at least in part, on a cloud server.

In the translation embodiment, a convolutional pose machine estimates the body pose, a region-based convolutional network finds the hands, and a second pose network gives the keypoints of the dominant and non-dominant hands. These are merged into one flattened feature vector per frame, normalized so that the head is at the origin and the shoulders are one unit apart, smoothed and appended to a feature queue. A one-dimensional convolutional network splits the queue into sign, transition and non-signing regions. A comparator first looks for the whole signed passage in a base of pre-recorded sentences using k-nearest neighbours with dynamic time warping; if none matches, each sign is matched in the same way against a base of single signs, and fingerspelled regions go to a separate sequence model. The resulting glosses are translated into English by a sequence-to-sequence grammar model and returned to the device as text. Other embodiments highlight the person who is signing in a video conference, answer signed questions,

and show an "ASL menu" of the three most likely sentences from the sentence base for the user to confirm (Fig. 11(a)). The granted claim 1 is broader than the published application: it claims capturing images, a first pose network for the body and a second for the hands, a flattened feature vector of fixed size, a feature queue, and its conversion into a target-language output, without requiring a server.

Difference from our proposal. Sign-Speak is the closest prior art. Both systems use one ordinary camera, body and hand pose features normalized to the body, sign splitting, a fingerspelling path and a list of candidates that the user confirms. Our proposal differs in four ways. First, Sign-Speak needs a network connection and sends the user's data to a server, while our device runs fully offline on an embedded GPU and no video leaves it. Second, Sign-Speak matches signs to stored examples by nearest neighbour, while we use a trained neural classifier whose output is calibrated as a probability. Third, Sign-Speak's candidate menu is a separate mode that always offers three whole sentences from a fixed bank, while our device decides for each sign, from its calibrated confidence, whether to accept it or to show five candidates, always asks for confirmation of the complete sentence before speaking, and measures this trade-off with risk–coverage curves (Fig. 8). Fourth, Sign-Speak returns English text, while our device speaks English, Turkish or Spanish and shows the hearing person's spoken reply as text. One caution follows from the review: because the granted claim 1 describes pose-based recognition with body and hand pose networks in general terms, a pipeline like ours, which uses the MediaPipe body and hand models and a sequence of flattened per-frame features, may fall within its scope even though the differences above are real. The differences establish the originality of the proposed system but do not by themselves avoid the claim, so a commercial version would need a freedom-to-operate review (Section 6.2).

Patent 2: Automated Sign Language Translation Using Multiple Input and Output Modalities (Avodah)

Table 20. Record of Patent 2.

Item	Details
Title	Automated sign language translation and communication using multiple input and output modalities
Number	U.S. Patent No. 10,489,639 B2, granted November 26, 2019 [18]; continuations include US 10,956,725 B2 and US 12,183,123 B2
Inventors / assignee	Michael Menefee, Dallas Nash, Trevor Chandler / Avodah, Inc. (originally Avodah Labs, Inc.)
Dates	Priority February 12, 2018; filed January 25, 2019
Legal status	Active; the four-year maintenance fee was paid in 2023
Link	https://patents.google.com/patent/US10489639B2/en

Abstract. "Methods, apparatus and systems for recognizing sign language movements using multiple input and output modalities. One example method includes capturing a movement associated with the sign language using a set of visual sensing devices, the set of visual sensing devices comprising multiple apertures oriented with respect to the subject to receive optical signals corresponding to the movement from multiple angles, generating digital information corresponding to the movement based on the optical signals from the multiple angles, collecting depth information corresponding to the movement in one or more planes perpendicular to an image plane captured by the set of visual sensing devices, producing a reduced set of digital information by removing at least some of the digital information based on the depth information, generating a composite digital representation by aligning at least a portion of the reduced set of digital information, and recognizing the movement based on the composite digital representation."

Description. The patent describes a device that lets a signer and a non-signer communicate in both directions. The device has a set of sensor inputs and outputs for each party; it identifies the language each person uses and outputs the translation in the form the other person prefers: text, audible speech, or a visual language such as glyphs, animations or a signing avatar, possibly in a different sign language. It can be used face to face (Fig. 11(b): sensors facing each person, a processor, a primary and a secondary display and a speaker) or by two parties in different places connected over a network. Translation can be performed "locally, remotely or both", with cloud resources used, for example, when many output sign languages must be supported.

The claims focus on how the signing is captured. Several visual sensors with multiple apertures view the signer from different angles, and a depth sensor, either structured light or time of flight, measures depth perpendicular to the image plane. The depth is used to discard part of the image data, for example the background, and the remaining views are aligned into one composite three-dimensional representation, from which a neural network recognizes the movement. Dependent claims add the capture of eyebrow movements, gaze direction and blinks, which carry grammatical information in sign languages. The patent has 30 claims, and Avodah holds a family of related patents on gesture recognition, training and user interfaces.

Difference from our proposal. Avodah's face-to-face embodiment is the closest to our device in layout: one unit between two people, with inputs and screens facing each side and a speaker (compare Fig. 11(b) with Fig. 2). The technical approach, however, is different. Every independent claim requires multiple camera apertures and a depth sensor, while our device uses one ordinary RGB camera and no depth sensor, which makes it cheaper and simpler and keeps it outside those claims. Avodah also captures facial grammar, which we do not, and outputs signing avatars and several sign languages, while our device recognizes ASL only and speaks three spoken languages. Offline operation is optional in Avodah's design and mandatory in ours, and Avodah does not describe abstaining or confirming before output, which are central to our design.

Patent 3: Translating Sign Language Content into Word Content and Vice Versa (Sorenson)

Table 21. Record of Patent 3.

Item	Details
Title	Methods, systems, and machine-readable media for translating sign language content into word content and vice versa
Number	U.S. Patent No. 12,131,586 B2, granted October 29, 2024 [39]
Inventors / assignee	John Carlock, Conrad Maxwell / Sorenson IP Holdings, LLC
Dates	Priority and filing April 9, 2021
Legal status	Active; Google Patents gives February 12, 2042 as the expected expiration
Link	https://patents.google.com/patent/US12131586B2/en

Abstract. "Communication systems, methods, and non-transitory machine-readable storage media are disclosed herein. A communication system may include a communication device

configured to receive a video stream including sign language content and any content indicators associated with the video stream during a real-time communication session within a single communication device or between a plurality of communication devices. The communication system may also include a translation engine configured to automatically translate the sign language content into word content during the real-time communication session without assistance of a human sign language interpreter. Further, the communication system may be configured to output the word content translation to a communication device during the real-time communication session."

Description. Sorenson provides video relay services, in which a Deaf person signs over video to an interpreter who voices the call to a hearing person. This patent describes replacing or assisting that interpreter with a translation engine. The engine divides the incoming video into sign language segments and compares each with stored video files, each linked to word content, and combines the matched words into a translation. It uses context in two ways: words already translated earlier in the session, and "content indicators", the facial expressions, body language and other visual cues that accompany the hand and arm movements and can change the meaning, express emotion or mark punctuation. The engine also works in reverse, producing sign language video from words.

Three configurations are described: a single device, such as a cell phone, that acts as a personal translator with an on-board translation engine; several devices connected over a network; and a video relay service in which servers translate ASL video into voice for the hearing party and the hearing party's voice into text for the Deaf user (Fig. 11(c)). In the relay configuration a training station, operated by a human who knows sign language or by an AI training engine, monitors sessions and updates the engine's databases to improve accuracy. The independent claim covers a communication device and a translation engine that translates each segment using the words already determined from earlier segments and the content indicators.

Difference from our proposal. This patent shows that a major provider of interpreting by video relay is itself pursuing automatic translation, which confirms the commercial interest in the field. Its setting, however, is a remote call through a relay service, while ours is a face-to-face conversation at a counter, where a relay service is not involved. It aims at continuous translation that uses context and facial cues, which is more ambitious than our recognition of single signs from

a fixed vocabulary, and it relies on servers and a monitoring trainer, while our device works alone and offline. The patent does not describe a confidence threshold or a confirmation step: the translation is passed to the other party during the session. In our device the Deaf user sees and confirms every sentence before it is spoken, and uncertain signs are offered as a list of candidates.

Patent 4: System and Method for Automated Sign Language Recognition (Bosch)

Table 22. Record of Patent 4.

Item	Details
Title	System and method for automated sign language recognition
Number	U.S. Patent No. 10,109,219 B2, granted October 23, 2018 [20]; published as application US 2016/0307469 A1; related EP 3284019 B1, CN 107690651 B, WO 2016/168591 A1
Inventors / assignee	Zhengyu Zhou, Tobias Menne, Kehuang Li, Kui Xu, Zhe Feng, Chin-Hui Lee / Robert Bosch GmbH
Dates	Priority April 16, 2015; filed April 15, 2016
Legal status	Active; the eight-year maintenance fee was paid in April 2026
Link	https://patents.google.com/patent/US10109219B2/en

Abstract. "A method for sign language recognition includes receiving, with an input device, an input based on a plurality of hand movements and postures of a user that correspond to a sequence of signs, extracting a plurality of features from the input corresponding to the plurality of hand movements and postures, identifying, a start of the sequence of signs in the input based on a first set of features in the plurality of features and a first Hidden Markov Model (HMM) stored in the memory and identifying a first sign in the input based on a second set of features in the plurality of features and a second HMM stored in the memory. The method also includes generating an output corresponding to the first sign from the input."

Description. The patent applies the methods of speech recognition to continuous sign language. Its input comes from "sensor signals from wearable device and/or image sequences from a camera", in its embodiments mainly digital gloves that measure the movements and angles of both hands. Each sign is modeled by a hidden Markov model, and the key idea is to model the movement between signs explicitly: separate models describe the transition before the first sign of a phrase, between two signs and after the last sign. The sign and transition models are connected into a decoding network, as words and pauses are in speech recognition, and a language model

gives the prior probability of each sentence, so that a whole phrase can be recognized without pauses between signs. The system treats the dominant and non-dominant hand separately; it can use only the global angles of the non-dominant hand, which allows a simpler glove on that hand, and it ties the non-dominant-hand parameters of one-handed signs to the transition model to reduce errors. The output device produces a text or audio translation of the recognized signs.

The granted claims are narrower than this general description. Independent claim 1 requires separate inputs for the movements of the dominant and the non-dominant hand, finger-related features extracted from the dominant hand and global-angle features from the non-dominant hand, a first hidden Markov model that detects the start of a sign sequence, a second one that identifies the first sign, and an output for that sign; the dependent claims add the transition and end models. The patent was also filed in Europe, China and through the Patent Cooperation Treaty, which shows that Bosch saw commercial value in sign language recognition as a human-machine interface, not only as an assistive aid.

Difference from our proposal. Bosch relies on wearable sensors, while our device needs nothing worn: the signer stands in front of one camera, which matches the preference of the Deaf users surveyed, of whom only 38% were comfortable with wearables [1]. Bosch uses hidden Markov models, while we use a neural network on body and hand landmarks. Its explicit transition models solve a problem we also face, recognizing several signs made without a pause; we currently handle it differently, with a rest pose between signs and, for fluent signing, candidate cuts where the hands slow down that the recognizer then scores (Section 7.3). Bosch's system is one-way and does not describe abstention, confirmation or multilingual speech.

Expired patent: Hitachi, U.S. Patent No. 5,887,069 [21]. Granted in 1999 to Hitachi, Ltd. with a priority date of 1992, this patent recognizes continuous sign language from data gloves or a camera by matching compressed time sequences against averaged reference patterns, and describes a translation system that also recognizes the signer's facial expression and adds the matching emotion to synthesized speech. Google Patents lists it as expired, so its methods are free to use. It shows that sign-to-speech translation has been patented for three decades; our device differs in using learned models, a single ordinary camera, abstention and confirmation, and several output languages.

8.4 Originality and Comparison

Table 23 compares the reviewed patents with the proposed device. Each feature of our design exists somewhere in the prior art: a single camera in Sign-Speak and Sorenson, a two-way device between two people in Avodah, user confirmation of candidate sentences in Sign-Speak, and several output languages in Avodah. No reviewed system, however, combines them, and none decides from a calibrated confidence when to accept a sign and when to abstain. The originality of our proposal therefore lies in the combination of fully offline processing on an embedded GPU with one ordinary camera, calibrated abstention followed by confirmation of every sentence before it is spoken, and multilingual speech with a reply channel in one stand-alone device; and in its evaluation, which reports accuracy for each signer and the trade-off between answering and abstaining. The review also showed the limits of this claim: recognition from body and hand pose is itself claimed by the granted Sign-Speak patent, and a major provider of relay services is working on the same problem. These findings are reflected in the legal constraints of Section 6.2.

Table 23. Comparison of the reviewed patents with the proposed device.

Feature	Sign-Speak [19]	Avodah [18]	Sorenson [39]	Bosch [20]	Hitachi [21]	Proposed device
Input	One camera	Several cameras and a depth sensor	Video from one device	Gloves (camera possible)	Data gloves or camera	One RGB camera
Where processing runs	Server required	Local, remote or both	Phone or relay servers	Local processor	Local computer	On the device, fully offline
Recognition method	Pose keypoints, nearest neighbour with time warping	Neural network on a 3D composite	Matching to stored videos with context	Hidden Markov models	Pattern matching	Pose landmarks, neural network
Calibrated confidence, abstains when unsure	Not described	Not described	Not described	Not described	Not described	Yes
User confirms before output	Yes, menu of three sentences	Not described	Not described	Not described	Not described	Yes, every sentence
Output	Text	Speech, text or avatar	Voice or text	Text or audio	Text or speech with emotion	Speech in EN, TR, ES
Reply from the hearing person	Not described	Yes	Yes, voice to text	Not described	Not described	Yes, speech to text
Legal status (Oct. 2026)	Active to 2043	Active	Active to 2042	Active	Expired	—

12. References

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